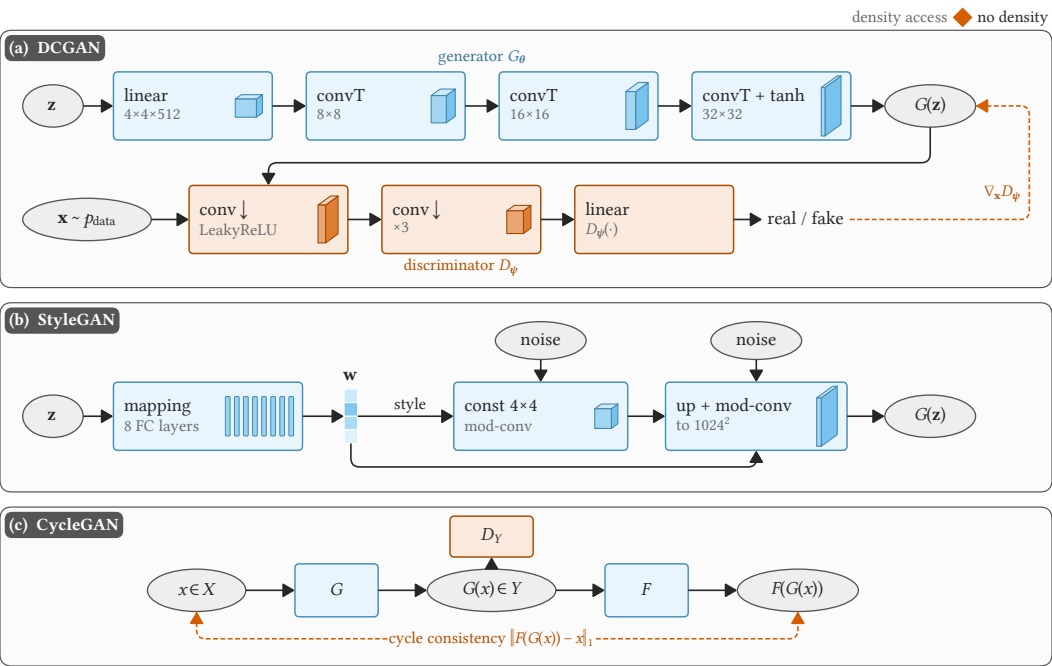
density access
no density
(a) DCGAN
generator G_θ
z
linear
4×4×512
convT
8×8
convT
16×16
convT + tanh
32×32
G(z)
$\nabla_x D_\psi$
x ~ p_{data}
conv ↓
LeakyReLU
conv ↓
×3
linear
$D_\psi(\cdot)$
real / fake
discriminator D_ψ
(b) StyleGAN
noise
noise
z
mapping
8 FC layers
w
style
const 4×4
mod-conv
up + mod-conv
to 1024^2
G(z)
(c) CycleGAN
D_Y
$x \in X$
G
$G(x) \in Y$
F
F(G(x))
cycle consistency $\|F(G(x)) - x\|_1$